Operating Systems Lab Report: Assignment 8

Memory Management in xv6

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Executive Summary

This consolidated report documents the design decisions, implementation details, memory utilization metrics, and debugging analysis for **Assignment 8: Memory Management in xv6** (Questions 1 through 4). Modern operating systems rely on virtual memory to achieve process isolation and efficient physical resource management. The four core memory management extensions implemented in this lab are:

1. **Virtual to Physical Address Translation (`va2pa`)**
2. **Physical Memory Statistics Instrumentation (`getmeminfo`)**
3. **Process Page Table Hierarchy Inspection (`vmprint`)**
4. **Demand Paging / Lazy Memory Allocation for the Heap (`sys_sbrk` & `T_PGFLT`)**

Section 1: Detailed Design Decisions and Key Data Structures

1.1 Question 1: Virtual to Physical Address Translation (`va2pa`)

- **Objective:** Expose kernel page table translation to user space and verify virtual-to-physical mapping mechanics.
- **Design & Implementation:**
    - Implemented system call `int va2pa(int va)` in `sysproc.c`.
    - Exposed `walkpgdir()` from `vm.c` in `defs.h` to look up the Page Table Entry (PTE) corresponding to address va. Called with allocation flag set to `0` for lookup only.
    - Validated that the PTE exists, the present bit (`PTE_P`) is set, and user-mode access (`PTE_U`) is granted. Returns `-1` if unmapped or inaccessible.

    - Extracted structural address components and printed them using `cprintf`

- **Testing (`vatest.c`):** Evaluated translations for a global variable, stack variable, and dynamic heap block (`malloc`). Verified that unmapped addresses such as $0x7FFFFFFF$ return `-1`. Called `fork()` and demonstrated that parent and child processes share identical virtual addresses for global variables but resolve to distinct physical frames.

1.2 Question 2: Physical Memory Statistics (`getmeminfo`)

- **Objective:** Instrument physical memory allocation to track free memory and per-process memory consumption in real time.
- **Data Structure (`struct meminfo`):**

```
C/C++
struct meminfo {
    uint total_pages;      // Total physical pages (PHYSTOP / PGSIZE)
    uint free_pages;       // Current free physical pages
    uint proc_pages;       // Calling process user pages (sz / PGSIZE)
    uint pagetable_pages;  // Pages allocated for process page tables
};
```

- **Design & Implementation:**
    - Added `freepages` counter to `struct kmem` in `kalloc.c`, synchronized using `kmem.lock`.
    - Updated `freepages` during every `kalloc()` (decrement) and `kfree()` (increment).
    - Traversed the process's page directory in `sys_getmeminfo()` to count valid non-null page directory entries and calculate `pagetable_pages`.
- **Testing & Leak Verification (`memtest.c`):** Logged baseline metrics, expanded heap memory via `sbrk()`, created a child process via `fork()`, and confirmed that `free_pages` returns exactly to its pre-fork baseline after the child exits and is reaped via `wait()`.

1.3 Question 3: Dumping the Process Page Table (`vmprint`)

- **Objective:** Visualize the hierarchical two-level page table structure for a running process.
- **Design & Implementation:**
    - Implemented `int vmprint(void)` in `vm.c` to traverse process page directory entries.
    - For each valid Page Directory Entry (PDE), outputs PDE index, physical address of the page table frame, and flags.

    - Traverses the 1024 Page Table Entries (PTEs) inside valid page tables, printing entry index, mapped virtual address, physical frame address, and flags (`P`, `W`, `U`).
    - Formatted output using indentation (`..` for level 1 PDE, `.. ..` for level 2 PTE).
    - Concludes with a summary line counting kernel-reserved entries.
- **Testing (`vmprinttest.c`):** Inspected page tables at startup, after `sbrk(8 * 4096)`, and after `sbrk(-4096)`. Verified that the stack guard page created by `exec()` has `PTE_U` cleared, confirming user-accessible pages match.

1.4 Question 4: Lazy Memory Allocation (Demand Paging)

- **Objective:** Defer physical page allocation during heap growth until pages are accessed by the program.
- **Design & Implementation:**
    - **System Call Modification:** Modified `sys_sbrk()` in `sysproc.c` so that positive heap requests ($n > 0$) expand `myproc()->sz` without calling `growproc()`. Heap reductions ($n < 0$) continue to deallocate physical pages immediately.

    - **Page Fault Trap Handler:** Updated `trap.c` to handle `T_PGFLT` traps. Read faulting virtual address from control register `cr2` using `rcr2()`.
    - Allocates a zeroed physical frame using `kalloc()`, maps it into the page table with `mappages(..., PTE_W | PTE_U)`, and returns to user space to re-execute the faulting instruction.

    - **Edge Cases & Kernel Safety:** Terminated processes attempting out-of-bounds access ($\geq sz$ or below stack) or when `kalloc()` fails. Updated `argptr()` and `copyin()` in `vm.c` to ensure kernel system calls touching untouched lazy buffers trigger page allocation on demand. Updated `copyuvm()` in `proc.c` to skip non-present PTEs during `fork()` instead of triggering kernel panic.

- **Testing (`lazytest.c`):** Allocated 32MB via `sbrk()`, touched distant pages, verified memory statistics, tested process duplication via `fork()`, and verified kernel safety during system calls using untouched buffers.

Section 2: Empirical Memory Usage Statistics (Q2 & Q4)

The table below summarizes physical memory metrics recorded across execution phases for Questions 2 and 4:

Execution State / Scenario	Total System Pages	Free Pages ($kmem$)	Process User Pages	Page Table Pages	Notes / Observations
Initial Boot Baseline (Q2)	32,768	32,210	3	3	Initial init/sh process setup
Post `sbrk(+10 Pages)` (Q2)	32,768	32,200	13	3	Eager physical frame allocation
Post `fork()` Creation (Q2)	32,768	32,187	13	4	Physical page duplication
Post Child Exit & Reap (Q2)	32,768	32,200	13	3	Restored; zero memory leaks
Post `sbrk(32 MB)` Lazy Call (Q4)	32,768	32,200	8,205	3	sz grows; 0 frames allocated
Post Touching 3 Lazy Pages (Q4)	32,768	32,197	8,205	3	Demand allocation of 3 frames

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Section 3: Problems Faced and Technical Resolutions

1. Kernel Panics on System Calls with Untouched Lazy Buffers

- **Problem:** Passing an unmapped lazy heap address to a system call (e.g., `read(fd, buf, len)`) caused a kernel panic because system call execution occurs in kernel mode, bypassing the standard user trap handler.
- **Resolution:** Updated kernel argument validation and memory copy functions (`argptr`, `copyin`) in `vm.c` to check if an address lies in valid lazy heap space. If unmapped, physical allocation and page table mapping are performed directly before dereferencing.

2. Header Structure Conflicts (`struct meminfo`)

- **Problem:** Redefining `struct meminfo` across multiple user files caused type redefinition conflicts and compiler warnings.
- **Resolution:** Consolidated `struct meminfo` strictly inside `user.h` and removed redundant local declarations across test programs.

3. Compiler Strictness (`-Werror`)

- **Problem:** xv6 build settings treat all compiler warnings as errors, causing build failures over formatting inconsistencies or unused prototypes.

- **Resolution:** Cleaned up single-line conditional statements with explicit curly braces `{}` and removed unused function declarations.

Conclusion

Questions 1 through 4 of **Assignment 8: Memory Management in xv6** have been fully implemented and verified. The implementation provides virtual-to-physical address translation (`va2pa`), real-time physical memory instrumentation (`getmeminfo`), process page table dumping (`vmprint`), and demand paging (`sys_sbrk` lazy allocation) with full kernel system call safety and zero memory leaks.